

Payments Guidelines Query Expert

Midterm Capstone Report | 4 October 2026
[Team ID] | [Team member names]

Problem and motivation

Payments teams must interpret requirements spread across circulars, products and revisions. Finding a relevant document is only the first step: users must identify the applicable conditions and check the evidence behind an answer. Manual searching takes effort, while an AI answer can sound convincing despite using the wrong source or omitting a condition.

Payments Guidelines Query Expert is a local AI assistant for product, operations and compliance professionals in banks and fintechs. It helps them ask a payments question, read a cited answer and inspect its source. Reduced research effort is the intended benefit.

Proposed solution and scope

Implemented capability	User outcome
Ask questions	Retrieve relevant passages and generate answers with citations.
Inspect guidelines	Search by product or reference, open source text and available original PDFs, and export notes.
Summarise and compare	Read selected documents together while preserving each circular’s attribution.

Initial corpus: NPCI circulars across multiple payment products. Preliminary evaluation focuses on UPI. Other payment networks, regulators and institution-specific guidelines are planned extensions.

Data and preprocessing

The imported register contains 1,921 records: 1,562 searchable documents and 359 unavailable or excluded entries. Searchable text spans 4,826 pages and 14,060 chunks. Markdown provides the text; product inventory JSON supplies identity, availability and provenance. This is a saved collection, not a live website feed.

A recognised heading or numbered clause starts a text block within a page. Long blocks use windows of at most 220 words with 35-word overlap. Chunks never mix circulars or cross page boundaries; each retains document ID, page, heading and neighbour links. BM25 indexes chunk text, Qwen embeddings go into Chroma, and SQLite stores metadata and relationships. Related circulars are linked, not merged.

System design and reasoning

Preparation and question flow

Preparation: Markdown and inventory → attributed chunks → keyword index, Chroma vectors and SQLite metadata.

Question flow: question and filters → relevant passages → bounded evidence → local answer generation → citation and support checks → answer with inspectable sources.

Module	Why it is included	Trade off to evaluate
Chunking and metadata	Keep passages focused and preserve document identity, pages and links.	Small chunks can miss conditions; extraction may be incomplete.
BM25 keyword search	Find exact terminology and circular references.	May miss a differently worded question.
Qwen embeddings and Chroma	Find passages with similar meaning.	Similar passages can still be irrelevant.
Hybrid retrieval and MiniLM reranking	Combine keyword and semantic results, then reorder candidates for focused questions.	Adds time; benefit over BM25 is not yet demonstrated.
Bounded context assembly	Fit selected passages and related evidence within model limits.	Relevant material may be omitted. Summaries and comparisons use section coverage and bypass reranking.
Qwen3.5 and support checks	Draft a readable answer; check citations, numbers and quoted support.	Same-model checks can repeat generation errors.
Source reader and export	Let users verify and reuse the answer with attribution.	A citation alone does not establish correctness.

Models and deployment

The frontend uses HTML, CSS and JavaScript with a Python backend. Ollama runs Qwen3.5 9B for generation and Qwen3 embedding 0.6B for semantic retrieval. MiniLM L6 v2 reranks locally through ONNX. SQLite stores metadata; Chroma stores vectors.

We use pretrained models without fine-tuning because the initial requirement is access to changing, attributable documents. The local machine has about 32 GB RAM. Generation uses a 16,384-token context, with a separate 18,000-character evidence budget and up to 3,072 output tokens.

Design decisions and evidence

The reasons above explain implementation choices, not proven superiority. The preliminary comparison on the next page tests retrieval behaviour. Independent answer review and controlled comparisons on harder questions are required before claiming that added complexity improves quality.

Progress evaluation and next steps

Implemented progress

The app supports cited answers, source inspection, summaries, comparisons and export. The slide demonstration asks about using Tap & Pay at a shop’s card machine and returns cited terminal conditions. Browser checks cover the core journeys and recovery; six earlier scenarios are also retained. A new question about sending money to a friend produced a general feature overview instead of a direct answer, showing why citation support alone is insufficient. Long summaries and slow comparisons remain improvement areas.

Preliminary retrieval comparison

Purpose: determine whether extra retrieval complexity improves evidence selection over a keyword baseline. Four methods ran the same frozen 50-case UPI set on the same corpus. The AI-assisted set contains 55 expected source passages and six negative cases; independent review is pending.

Retrieval method	Expected evidence found	Negatives with no result	Median search ms
BM25	55/55	6/6	12.0
Vector	55/55	5/6	83.0
Hybrid	55/55	5/6	78.0
Hybrid + reranker	55/55	5/6	160.5

Some questions need multiple passages: 50 cases therefore contain 55 expected evidence matches. All methods found 55/55. The negative-case column checks whether search returns no passages when none are expected: BM25 did so in 6/6 cases; semantic methods returned irrelevant passages in one. Search time is the median retrieval time and excludes answer generation.

Inference: BM25 is a strong baseline on this set. Reranking raised hybrid median search time from 78 to 160.5 ms without improving measured coverage. This is not an answer-accuracy score or proof that BM25 is always best. Many queries name circulars; 15 summary/comparison routes bypass reranking. Timing is one preliminary warmed pass. Harder, independently reviewed questions are needed before selecting a method on quality grounds.

Next steps and completion evidence

Indicative timing	Implementation work	Completion evidence
Weeks 1–2 after midterm	Independently review the 50 cases and freeze unseen questions.	Reviewed test set and answer-quality scorecard: correctness, citations, omissions and abstention.
Weeks 3–5	Repair source and answer failures, shorten long summaries, compare retrieval settings.	Tested revision and a documented quality versus latency comparison.
Weeks 6–8	Observe payments users on research tasks; prepare repeatable setup and final demo.	Task outcomes, corrections and time; a reproducible local demonstration.

Reviewer access and source completeness are dependencies. A second payments source is optional after validation, beyond the core completion gate. Feedback requested: evaluation priorities and evidence of user value.